Filtering Through the Storm

Discussion of

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The views expressed are my own and do not necessarily represent those of the Federal Reserve Bank of Kansas City or the Federal Reserve System.

Main Idea

- How should a DSGE model handle episodes of extreme volatility?
- Authors conduct a horse race of four treatments, in Colombian model (CUMBIA)
  1. Constant variance shocks.
  2. Treat real activity variables as missing during volatile episodes
  3. Add episode-specific AR(1) processes
  4. Add episode-specific white noise
- Conclude T3 forecasts best post-COVID; also favor T4 for implementation/explanatory ease

Difference between Treatments 3 and 4 (paper's equations)

Let $x_t \equiv \zeta_t - \bar{\zeta}$:

Treatment 3, Eq. (4.1): $x_t = \rho x_{t-1} + \varepsilon_t + \varepsilon_t^H$

Treatment 4, Eq. (4.3): $x_t = \rho x_{t-1} + \varepsilon_t + \varepsilon_t^H - \rho \varepsilon_{t-1}^H$

Difference between Treatments 3 and 4 (lag operator notation)

Lag operator $Lx_t = x_{t-1}$:

Treatment 3: $(1 - \rho L)x_t = \varepsilon_t + \varepsilon_t^H$

Treatment 4: $(1 - \rho L)x_t = \varepsilon_t + (1 - \rho L)\varepsilon_t^H$

Difference between Treatments 3 and 4 (invert)

For $|\rho| < 1$,

$$\text{Treatment 3:} \qquad x_t = \frac{1}{1-\rho L}\varepsilon_t + \frac{1}{1-\rho L}\varepsilon_t^H$$

$$\text{Treatment 4:} \qquad x_t = \frac{1}{1-\rho L}\varepsilon_t + \varepsilon_t^H$$

Additional term is AR(1) with same ρ under T3; white noise under T4.

Major Comment 1: Real-time Forecasting

The current exercise is "pseudo out-of-sample" in that:

- Each date's forecast Kalman filters using data up to that date
- ... but uses the parameter values from the entire 2000Q1–2025Q4 sample's MLE.
- Especially problematic for forecasting at the beginning of an episode

I recommend a more complete out-of-sample test, with the model re-estimated at each date, using only data available up to that point.

Forecast rankings depend on when we ask

Average RMSE improvement relative to Treatment 1, across GDP growth, inflation, and the policy rate:

	Pre-COVID	During COVID	Post-COVID
Treatment 2: omit real data	−6.6%	−15.2%	−8.6%
Treatment 3: persistent	−2.3%	−9.1%	+12.8%
Treatment 4: transitory	−5.1%	−11.2%	+6.9%

T1 is best before/during COVID, T3 is best afterwards. Authors should make clear why they care more about the latter. Say both:

- crisis quote here
- post-crisis quote here

Major comment 2: What exactly is T4's advantage?

Treatment 3 forecasts better than Treatment 4 in every row of Table 6.1. So why argue for using both? Need to clarify:

- T4 is easier to explain to policy makers?
- T4 is easier to implement?
  - Authors mention some concerns about collinearity in T3
  - So, is it identified or isn't it?
- Explain the choice of $s_\tau = (2, 2, 3)$ and show forecast sensitivity to it. What operational benefit makes Treatment 4 worth its lower forecast accuracy?

Major comment 3: Why Treatment 2?

- Plenty of citations for the other methods; do people actually use T2?
- Clarify: Why is it a reasonable idea to discard several time series?
- .... and why these seven: GDP, consumption, investment, exports, imports, foreign GDP, and trend growth

Ideas for increasing potential impact

Practioners will wonder: do the CUMBRIA conclusions generalize?

- Repeat the horse race in other DSGE models (many official central bank models are available)
- Can we expand the horse race? i.e. other approaches for non-Gaussianity:
  - peristent stochastic volatility
  - fat-tailed innovations
- Extended sample? 100 quarters + 27 observables for 50+ paramters would beenfit from more data
- Additional settings? CUMBRIA appears general enough to apply to other small open economies.

Extra thoughts

- Non-Gaussianity vs non-linearity
- AI use: writing (clarity vs strategy), ambition